



TRIBHUVAN UNIVERSITY
INSTITUTE OF ENGINEERING
NATIONAL COLLEGE OF ENGINEERING

A
FINAL PROJECT REPORT
ON
**EDUMENTX: LOCATION BASED TUTOR DISCOVERY
PLATFORM**

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Abstract

Finding a qualified home tutor in the Kathmandu Valley remains a fragmented, broker-dependent process. Students and parents rely on word-of-mouth referrals, social media groups, or local advertisements, resulting in mismatched expectations, wasted time, and limited trust in tutor qualifications. Existing tutoring platforms are either online-class-focused or lack location-based discovery, leaving a gap for in-person home tutoring services. EdumentX is a mobile application built with React Native and Expo that connects students and parents with verified home tutors in their locality. The platform provides map-based tutor discovery using Google Maps, a distance-based ranking system that promotes tutors by proximity and subscription tier, and a Retrieval-Augmented Generation (RAG) chatbot powered by Groq that answers natural-language queries about available tutors. Tutors undergo an AI-assisted identity verification pipeline that extracts text from citizenship documents using Groq Vision OCR and validates profile photographs, with automatic approval for high-confidence matches and manual admin review for ambiguous cases. The system supports enrollment management, group batch scheduling, in-app messaging, ratings and reviews, and an eSewa-based Pro subscription model for tutors. Firebase Authentication and Cloud Firestore provide identity and data persistence, while Supabase handles object storage and serverless edge functions. Security rules enforce owner-only data access and role-based permissions. The application runs on Android with Google Maps integration and is designed to operate within free-tier quotas, making it viable as a zero-cost demonstration platform. Testing includes 47 unit tests for enrollment logic, 21 lint rule tests for design token enforcement, and six Firestore security rules emulator suites. The results demonstrate that a location-first, trust-oriented tutoring marketplace can be built and deployed without paid infrastructure at demo scale.

Keywords: Location-Based Services, Tutor Discovery, React Native, Firebase, KNN Recommendation, RAG Chatbot, AI Verification

Contents

Page of Approval	ii
Copyright	iii
Acknowledgements	iv
Abstract	v
List of Figures	vi
List of Tables	vii
List of Abbreviations	ix
1 Introduction	1
1.1 Background	1
1.2 Problem Statement	2
1.3 Objectives	3
1.4 Scope	3
2 Literature Review	5
2.1 Related Work	5
2.2 Related Theory	6
2.2.1 Location-Based Services	6
2.2.2 Haversine Distance Formula	6
2.2.3 Geofencing and Radius-Based Filtering	7

2.2.4	K-Nearest Neighbors Algorithm	7
2.2.5	Retrieval-Augmented Generation	7
2.2.6	React Native and Expo	8
2.2.7	Firebase and Supabase	8
2.2.8	Google Maps via expo-maps	8
3	Methodology	9
3.1	System Design	9
3.1.1	System Architecture	9
3.1.2	Use Case Diagram	10
3.1.3	Data Flow Diagram	12
3.1.4	RAG Chatbot Workflow	13
3.2	Implementation Details	14
3.2.1	Frontend Development	14
3.2.2	Backend Development	14
3.2.3	Testing Strategy	14
3.3	Module Implementation	15
3.3.1	Onboarding and Authentication Module	15
3.3.2	Tutor Profile and Verification Module	15
3.3.3	Tutor Discovery and Map Module	15
3.3.4	Enrollment and Batches Module	15
3.3.5	Messaging Module	16
3.3.6	Pro Subscription Module	16
3.3.7	Admin Module	16
4	System Design	17
4.1	Requirement Specification	17

4.1.1	Functional Requirements	17
4.1.2	Non-Functional Requirements	18
4.1.3	System Architecture	19
4.1.4	Use Case Diagram	19
5	Results and Discussion	20
5.1	Software Development Approach	20
5.2	Testing and Evaluation	20
5.2.1	Unit Test Results	20
5.2.2	Lint Rule Test Results	20
5.2.3	Security Rules Test Results	21
5.2.4	Build Verification	21
5.3	Current Build	21
6	Conclusions	23
7	Limitations and Future Enhancement	24
7.1	Limitations	24
7.2	Future Enhancement	24
8	References	26

List of Figures

3.1	System Architecture of EdumentX	9
3.2	Use Case Diagram	11
3.3	Level 0 Data Flow Diagram	12
3.4	Level 1 Data Flow Diagram	13
3.5	RAG-Based AI Chatbot Workflow	13
5.1	EdumentX Current Build on Android	22

List of Tables

5.1	Unit Test Coverage Summary	20
5.2	ESLint Rule Test Results	21
5.3	Security Rules Test Suites	21

List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
DFD	Data Flow Diagram
GPS	Global Positioning System
HMAC	Hash-based Message Authentication Code
HTTP(S)	Hypertext Transfer Protocol (Secure)
IOE	Institute of Engineering
JSON	JavaScript Object Notation
KNN	K-Nearest Neighbors
LBS	Location-Based Service
LLM	Large Language Model
NCE	National College of Engineering
NPR	Nepalese Rupee
OCR	Optical Character Recognition
RAG	Retrieval-Augmented Generation
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SEE	Secondary Education Examination (Grade 10, Nepal)
TU	Tribhuvan University
UI	User Interface
UX	User Experience

1. Introduction

1.1 Background

Education is one of the most important foundations for personal growth and social progress. Around the world, families invest heavily in their children's education, recognizing that formal schooling alone may not always be enough. When students struggle with certain subjects or need extra preparation for exams, many families turn to private tutoring as a way to fill the gaps. This practice of hiring tutors outside of regular school hours has become extremely common, as parents look for additional academic support to help their children perform better.

Private tutoring, commonly referred to as shadow education, has become a widespread practice as formal schooling systems often fail to meet the diverse learning needs of students [1]. This form of supplementary education has grown particularly fast in developing countries, where public school resources are limited and teacher-student ratios remain inadequate [2]. Research on shadow education as a global phenomenon confirms that it has expanded across all continents and income levels, becoming a parallel curriculum that operates alongside formal schooling [3].

Nepal's education system has made notable progress over the past decade. According to the National Population and Housing Census 2021, the overall literacy rate reached 76.3 percent among the population aged five years and above, up from 65.9 percent in 2011. The country has also experienced rapid urbanization during this period, with the urban population share jumping from 17.07 percent in 2011 to 66.17 percent in 2021 [4]. Despite these improvements, challenges within the mainstream education system persist. Overcrowded classrooms, inconsistent teaching quality, and limited resources in public schools continue to push families toward private tutoring. In cities like Kathmandu, dual-income households have become increasingly common, leaving working parents with limited time for direct academic supervision of their children. As a result, many urban families look for home tutors as a practical solution.

The phenomenon of shadow education is deeply rooted in Nepal's educational landscape. Thapa's study on private and public school competition in Nepal highlights how supplementary tutoring has become an established practice, with families across income levels

relying on it to fill gaps left by formal schooling [5]. Studies further indicate that approximately 68 percent of Grade 10 students in Nepal receive some form of private tutoring, confirming how widespread this practice has become at the secondary level.

Despite this growing demand, Nepal's tutoring market remains fragmented and informal. Parents and students typically rely on word-of-mouth recommendations or unstructured social media groups to find tutors. These methods offer no verification of tutor credentials, no proximity-based filtering, and no intelligent matching. With smartphone penetration in Nepal reaching 65 to 71 percent, the digital infrastructure exists to address these problems through a dedicated mobile platform.

This report presents EdumentX, a location-based mobile application designed to modernize home tutor discovery in Kathmandu Valley. The platform enables students and parents to find nearby verified tutors through an interactive map, enroll in long-term monthly plans, and receive personalized recommendations through an AI-powered RAG chatbot.

1.2 Problem Statement

The process of finding a home tutor in Nepal is currently slow, unreliable, and frustrating for everyone involved. Both students seeking academic help and tutors looking for teaching opportunities face significant difficulties due to the lack of any organized system.

The core problems are:

- (i) There is no way for parents to check whether a tutor is actually qualified or trustworthy before hiring them. Most families rely on recommendations from people they know, which limits their options and gives no real measure of teaching quality or safety.
- (ii) Independent tutors, including university students looking for part-time work, retired teachers, and subject specialists, have no proper platform to advertise their services. Many end up going through agencies that take large commissions, while others simply cannot find students outside their own neighborhood.
- (iii) The tools people currently use to find tutors, mainly WhatsApp groups and Facebook pages, are not designed for this purpose. Posts are unorganized, location details are vague (just area names with no actual distance), there is no way to filter or search, and there is no system to verify anyone's identity or qualifications.

At present, no mobile application exists in Nepal that lets users find tutors on a map, filter by distance, or get AI-based suggestions. EdumentX aims to solve these problems by offering proximity-based search on an interactive map, a verification system where admins review tutor documents and award Blue Tick badges, a structured monthly enrollment process, and a chatbot that pulls live tutor data to give personalized recommendations.

1.3 Objectives

- To develop a location-based mobile application that enables students and parents to discover and connect with nearby home tutors in Kathmandu Valley using map-based proximity search.
- To integrate an AI-powered RAG chatbot that allows users to find suitable tutors through natural language queries grounded in live platform data.
- To implement a tutor verification system that helps parents make safe and informed hiring decisions.
- To implement structured enrollment management, group batch scheduling, in-app messaging, and a Pro subscription model for tutors.

1.4 Scope

EdumentX is a role-based mobile platform for connecting students and parents with home tutors based on location, subject, and monthly enrollment preferences. The scope of this project includes the following:

- **Target Users:** Students, parents, independent home tutors, and platform administrators.
- **Geographic Focus:** Kathmandu Valley, where smartphone penetration and internet connectivity are sufficient to support the platform.
- **Service Type:** Physical (in-person) home tuition services only; no online or video-based classes.
- **Academic Levels:** School and college students at SEE, +2, and undergraduate levels.
- **Platform:** Android application built with React Native (Expo SDK 54) and NativeWind 4.2.
- **Discovery Methods:** Map-based search with proximity ranking and an AI chatbot for natural language queries.

- **Enrollment Model:** Long-term monthly enrollment plans managed through structured request and response flows, with slot-based scheduling.
- **Verification:** AI-assisted identity verification with automatic approval for high-confidence matches and manual admin review for ambiguous cases. Blue Tick badge system.
- **Group Learning:** Support for group tuition batches created by tutors for shared, cost-effective learning.
- **Subscription:** Pro Tutor subscription via eSewa payment gateway (sandbox) for enhanced visibility and features.

2. Literature Review

2.1 Related Work

This section reviews recent academic research on location-based tutor matching systems, AI-powered educational platforms, and identifies the gaps that EdumentX aims to address.

A 2022 study presented at ASCILITE introduced MENTOR, an intelligent mobile peer tutoring application designed for both face-to-face and remote tutoring [6]. The system focused on matching peer tutors with students based on subject expertise and availability. While it demonstrated the value of mobile platforms for tutoring, it did not include any location-based features such as map visualization or proximity ranking. The matching was based solely on academic parameters without considering geographic distance between users.

A 2026 study published in the Journal of Emerging Technologies and Innovative Research described a Flutter-based mobile application for finding local tutors in India [7]. The system used Google Maps API for proximity-based tutor search and Firebase as the backend database. Students could filter tutors by subject, availability, and location. The application calculated distances between users using the Haversine formula and included features for booking and rating tutors. However, the researchers noted that AI-based recommendations were left as future scope. The entire matching process relied on manual filtering by the user, with no intelligent ranking or conversational search capability.

In 2025, researchers from Malaysia developed iTeachU, a system that combined AI with geolocation for tutor matching [8]. The platform used content-based filtering along with the K-Nearest Neighbors algorithm, incorporating Haversine distance calculations to rank tutors by both preference match and physical proximity. It was built using Flutter for the frontend and Flask for the backend. While the system demonstrated effective ranking within the Melaka region, its AI component was entirely rule-based. Users could not interact with the system using natural language. Instead, they had to select from predefined filters and dropdown menus, making the experience less flexible than a conversational interface.

All of these studies represent important steps toward technology-driven tutor discovery, but several gaps remain when considering the Nepali context. No existing research or

platform provides geospatial visualization of tutors on interactive maps with service area coverage for Nepal. City-level dropdown menus, which are used by most platforms, are not useful in dense urban areas like Kathmandu where a tutor two kilometers away is very different from one ten kilometers away. Independent part-time tutors such as university students and retired teachers lack a proper digital channel to reach nearby students, as existing systems tend to favor agency-managed tuition centers. None of the reviewed systems offer a natural language interface where users can simply type what they need instead of navigating through multiple filter options. While some platforms claim AI-based matching, none combine proximity, qualifications, availability, and budget into a single intelligent ranking powered by a large language model. The current methods available in Nepal, primarily social media groups, require significant manual effort to browse posts, contact agencies, and compare options without any automated assistance.

EdumentX brings map-based tutor visualization, a conversational interface for natural language queries, a structured verification system with Blue Tick badges, and a focus on empowering independent tutors to connect with nearby students efficiently.

2.2 Related Theory

2.2.1 Location-Based Services

Location-Based Services deliver context-aware services using a device’s real-time geographic position obtained through GPS, cellular triangulation, or Wi-Fi. In EdumentX, location-based services power the proximity-based tutor discovery and map visualization features. Tutor coordinates are stored as Firestore GeoPoint fields, which allow geospatial queries to be executed directly from the client application.

2.2.2 Haversine Distance Formula

The Haversine formula computes the great-circle distance between two points on Earth’s surface, accounting for the planet’s curvature. EdumentX uses this formula to rank tutors by their true distance from the student. The computation proceeds as follows:

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2\left(\frac{\Delta\lambda}{2}\right) \quad (2.1)$$

$$c = 2 \arctan 2(\sqrt{a}, \sqrt{1-a}), \quad d = R \cdot c \quad (2.2)$$

where ϕ_1, ϕ_2 are latitudes in radians, λ_1, λ_2 are longitudes in radians and $\Delta\phi$ and $\Delta\lambda$ are

the coordinate differences, $R \approx 6,371$ km, and d is the output distance. Degree values from GeoPoints are converted using $\phi_{\text{rad}} = \phi_{\text{deg}} \times \frac{\pi}{180}$. For the 1 to 20 km range typical in Kathmandu Valley, this formula provides meter-level accuracy.

2.2.3 Geofencing and Radius-Based Filtering

Before applying the Haversine formula, EdumentX uses a rectangular bounding box pre-filter to narrow down the candidate set and reduce computation [9]. For a given search radius r and student position (ϕ_s, λ_s) :

$$\Delta\phi_{\text{max}} = \frac{r}{R} \cdot \frac{180}{\pi}, \quad \Delta\lambda_{\text{max}} = \frac{r}{R \cdot \cos(\phi_s)} \cdot \frac{180}{\pi} \quad (2.3)$$

Only tutors whose coordinates fall inside this bounding box proceed to the exact Haversine distance check, which keeps query response times low.

2.2.4 K-Nearest Neighbors Algorithm

KNN is a non-parametric algorithm used in EdumentX to rank tutors by proximity. The primary ranking criterion is Haversine distance, with Pro subscription tier as a secondary sort key that promotes subscribers above non-subscribers within the same distance band. The default search radius is 15 km, which covers the Kathmandu Valley extent.

2.2.5 Retrieval-Augmented Generation

Retrieval-Augmented Generation is an approach that combines information retrieval with text generation to produce responses grounded in actual data rather than relying solely on a language model’s training data. The original RAG framework was introduced by Lewis et al., who showed that combining a pre-trained retriever with a sequence-to-sequence generator significantly improves accuracy on knowledge intensive tasks [10]. In EdumentX, RAG is used to retrieve matching tutor profiles from Firestore based on the user’s query and inject them as context into the Groq API (Llama 3 model) prompt, ensuring that recommendations are based on real, live platform data. A recent survey on RAG chatbots in education confirms that this approach reduces hallucination and improves the relevance of generated responses in educational contexts [11].

2.2.6 React Native and Expo

The EdumentX mobile app is built with React Native, which lets a single codebase produce native iOS and Android binaries. Expo sits on top of React Native and provides a managed toolchain for device features such as location, camera, and push notifications, along with over-the-air updates that shorten the release cycle.

2.2.7 Firebase and Supabase

Four services form the EdumentX backend. Firebase Authentication handles email/password and Google sign-in. Cloud Firestore is the main NoSQL store; tutor locations are saved as GeoPoint fields, and Security Rules control who can read or write which documents. Supabase Storage (1 GB free) holds verification documents and profile photos. Supabase Edge Functions provide serverless compute for the RAG chatbot (Groq API), eSewa payment signing, and AI identity verification. The Firebase Admin SDK is used out-of-band from a local Node.js script to grant the administrator role to a fixed list of emails.

2.2.8 Google Maps via expo-maps

EdumentX uses Google Maps rendered natively through the expo-maps module on Android, using the device's native Google Play Services. Custom teardrop pins with tutor avatars are rasterized offscreen and loaded into native map markers. The Google Maps API key is restricted to the Android app's package name and SHA-1 fingerprint.

3. Methodology

This chapter describes the methodology implemented for problem-solving, including the feasibility study, system design, implementation details, and module-level descriptions of every feature in the EdumentX platform.

3.1 System Design

3.1.1 System Architecture

Figure 3.1 shows the overall system architecture of EdumentX. The architecture has three layers: the client side, the serverless backend, and external services.

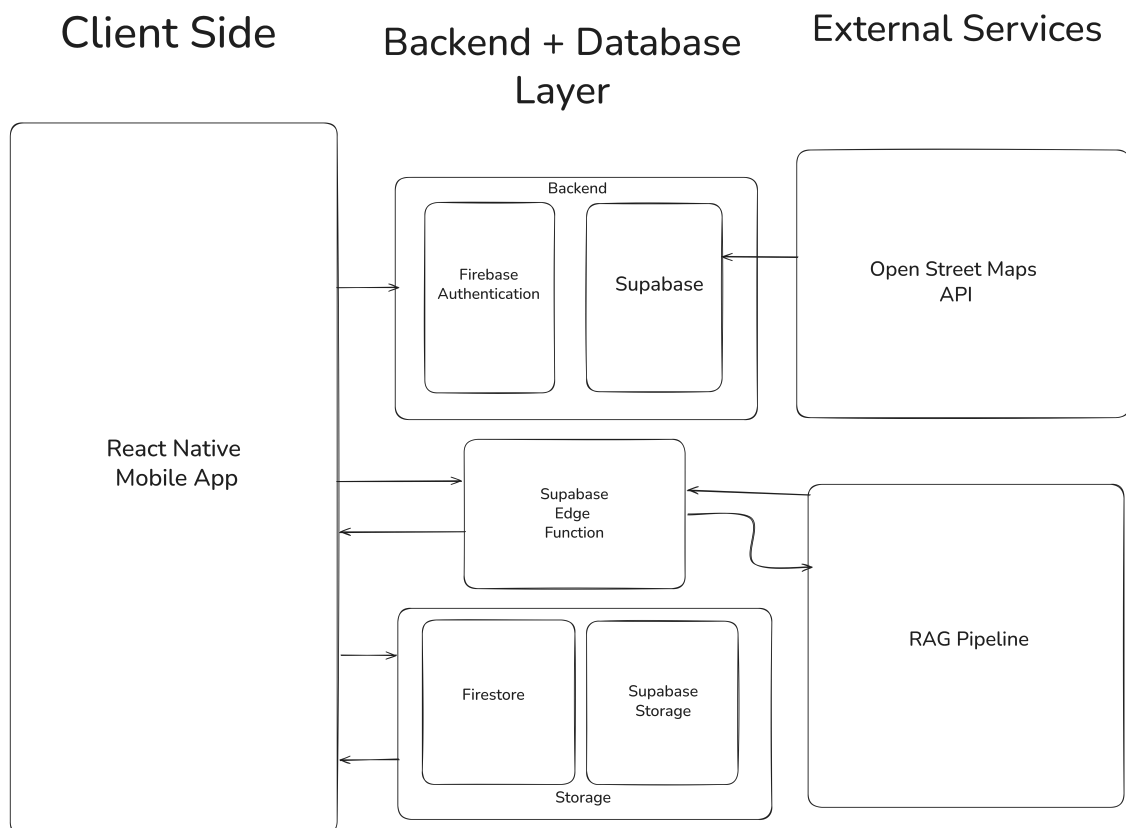


Figure 3.1: System Architecture of EdumentX

The client side is the React Native mobile application built with Expo. It serves as the sin-

gle entry point for all users and communicates with the backend using the native Firebase SDK.

The serverless backend runs across two platforms. Firebase provides Authentication (email/password + Google sign-in) and Cloud Firestore for structured data with Security Rules. Supabase provides object storage (1 GB free) and Edge Functions for the RAG chatbot (Groq API), eSewa payment signing, and AI identity verification (Groq Vision). Firebase Cloud Functions are scaffolded for Firestore-centric scheduled work.

The external services layer includes Google Maps (via expo-maps native module on Android), the Groq API for LLM inference (chatbot and vision), and Nominatim for keyless geocoding.

3.1.2 Use Case Diagram

Figure 3.2 shows the use case diagram with three actors: Students/Parents, Tutors, and Admin.

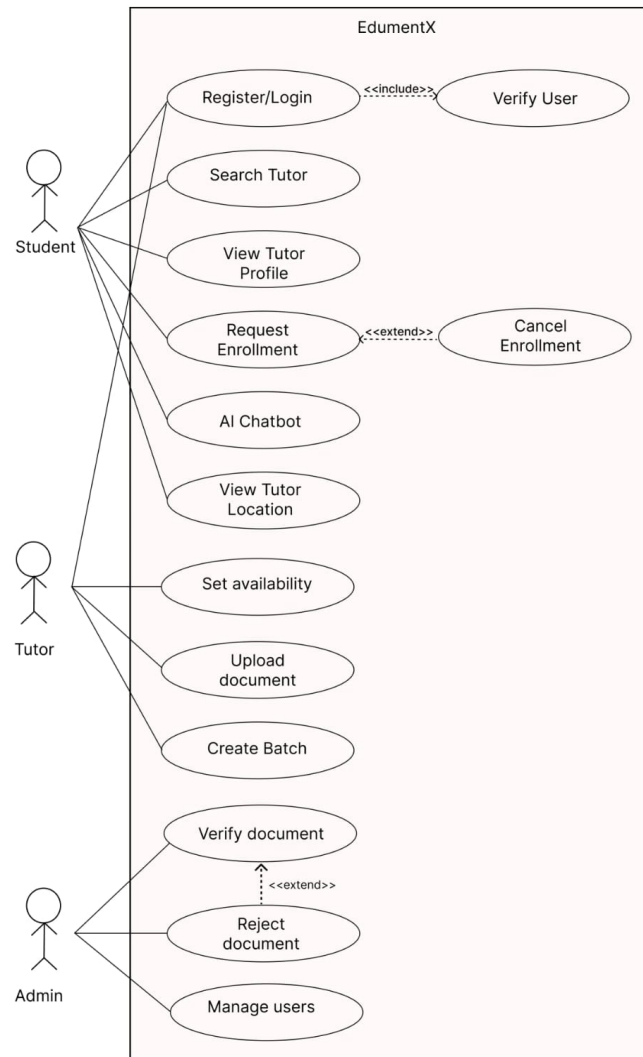


Figure 3.2: Use Case Diagram

Students can register, sign in, search for tutors using filters or the AI chatbot, view tutor profiles, send enrollment requests, join group batches, message tutors, save preferred tutors, and rate tutors.

Tutors can register, set their availability, upload documents for verification, manage their profile, respond to enrollment requests, create group batches, view their dashboard with analytics, and upgrade to a Pro subscription.

Admin can log in to the admin panel, review and approve/reject tutor verifications (with AI pre-screening), manage user accounts, and monitor platform statistics.

3.1.3 Data Flow Diagram

Figure 3.3 shows the Level 0 Data Flow Diagram.

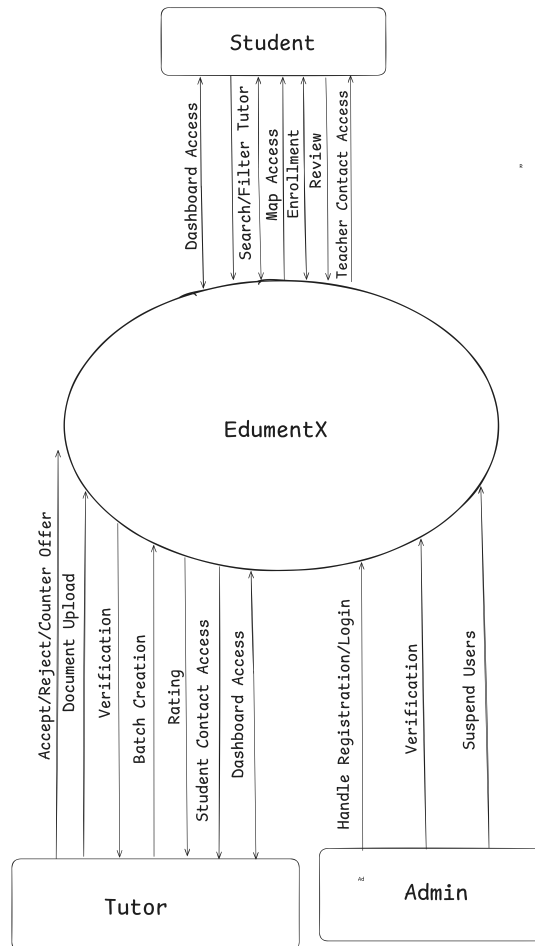


Figure 3.3: Level 0 Data Flow Diagram

Figure 3.4 shows the Level 1 Data Flow Diagram with detailed system processes including authentication, tutor search, AI chatbot recommendation, enrollment management, tutor verification, and rating handling.

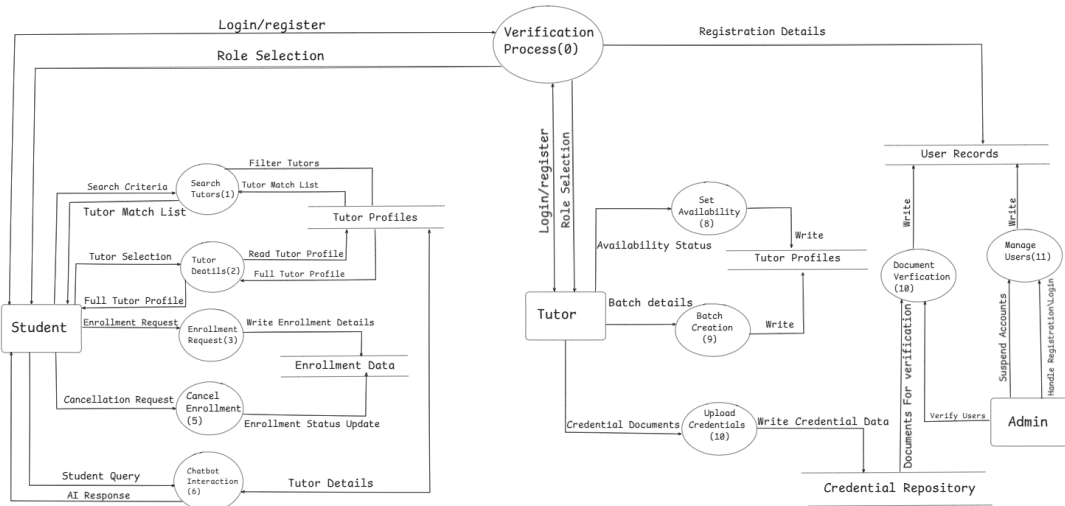


Figure 3.4: Level 1 Data Flow Diagram

3.1.4 RAG Chatbot Workflow

Figure 3.5 illustrates the RAG workflow used by the EdumentX AI chatbot.

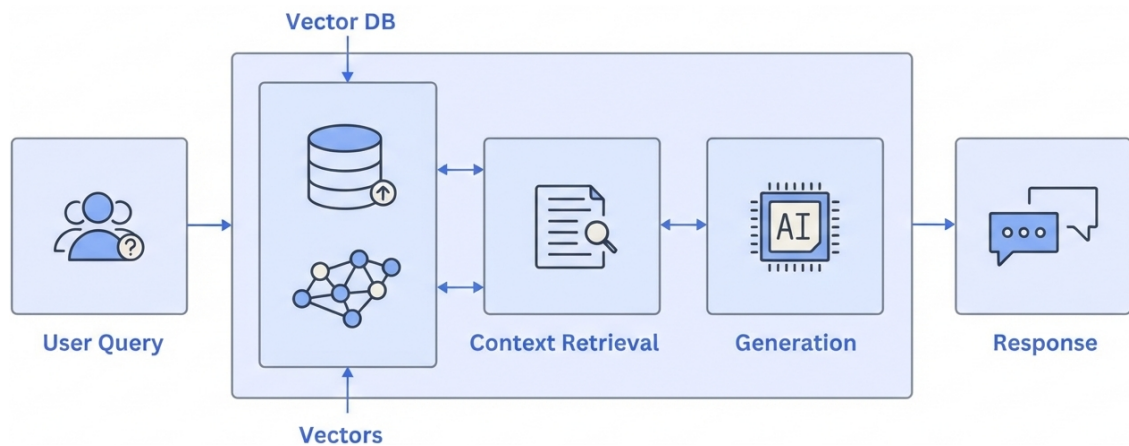


Figure 3.5: RAG-Based AI Chatbot Workflow

The chatbot works through a six-step pipeline. First, the student types a natural language query like “I need a science tutor near Lalitpur for 3 days a week under Rs 5,000.” The full conversation history is included so the chatbot can handle follow-up questions. Second, the client extracts structured parameters from the query: subject, location, budget, days per week, and plan duration. Third, the client queries the Firestore tutors collection, applies the bounding-box pre-filter followed by exact Haversine distance, and applies budget and subject filters. Fourth, the top ten matching tutor profiles are formatted and injected into the Groq API prompt as context data. Fifth, the language model generates a

response recommending three to five tutors with name, rate, distance, rating, and verification status. Sixth, the response is rendered as a chat message in the app.

3.2 Implementation Details

3.2.1 Frontend Development

The mobile application is built with React Native (v0.81.5) using TypeScript (v5.9.2) and Expo SDK 54. Styling is handled through NativeWind 4.2, which maps Tailwind CSS classes to React Native styles via a custom Babel plugin. Navigation uses Expo Router v6, a file-based router where each file in `src/app/` automatically becomes a route. The design system uses a custom color palette defined in `tailwind.config.js` with tokens such as `bg-night`, `text-amber`, `bg-surface`, and `border-border`. Five custom ESLint rules enforce design token compliance.

3.2.2 Backend Development

The backend is a hybrid architecture. Firebase provides Authentication and Cloud Firestore with Security Rules. Supabase provides object storage and three Edge Functions: `chat` (RAG chatbot using Groq), `create-esewa-order` (eSewa payment signing and verification), and `verify-identity` (AI document verification using Groq Vision). The `dataSource` selector pattern allows switching between mock data repositories and Firebase/Supabase repositories at runtime.

3.2.3 Testing Strategy

Testing is conducted at three levels:

1. **Unit tests** (`npm run test:derived`): 47 tests covering every pure helper function in `services/enrollments/derived.ts` and `types.ts`.
2. **Lint rule tests** (`npm run test:lint-rules`): 21 tests verifying the five custom ESLint rules that enforce design token compliance.
3. **Security rules tests** (`npm run test:rules`): Six emulator suites exercising Firestore security rules via the REST API.

3.3 Module Implementation

3.3.1 Onboarding and Authentication Module

The onboarding flow begins with a splash screen displaying the Edu-mentX logo, followed by a three-screen carousel introducing the platform's value propositions. Authentication is handled by a single screen (`EmailSignUp.tsx`) that hosts both sign-up and log-in forms. Email registration calls `createUserWithEmailAndPassword` followed by `sendEmailVerification`. Google Sign-In uses `@react-native-google-signin/google-signin`. The auth-flow routing in `_layout.tsx` runs a 5-step redirect tree based on authentication state, email verification, and role assignment.

3.3.2 Tutor Profile and Verification Module

Tutors complete a multi-step profile form. The AI verification pipeline triggers automatically on profile submission. The `verify-identity` Supabase Edge Function performs two checks using Groq Vision (Qwen 3.6 model): OCR extraction of the English name from the citizenship certificate, and face detection on the profile photograph. If both checks pass with confidence ≥ 0.9 , the function auto-approves the tutor by writing to `tutorVerifications/{uid}` via the Firestore REST API, updating the profile's `verificationStatus`, and creating the `tutors/{uid}` discovery document. Low-confidence results are routed to manual admin review.

3.3.3 Tutor Discovery and Map Module

The student home screen displays nearby tutors ranked by Haversine distance with Pro subscription priority. The `MapSearch` screen renders Google Maps via `expo-maps` with custom teardrop pins showing tutor avatars. A filter sheet allows students to adjust distance radius, budget range, subject, grade level, and tutoring mode. Tutor cards display name, photo, headline, subjects, rating, monthly rate, distance, and verification badge.

3.3.4 Enrollment and Batches Module

Students send enrollment requests to tutors. Requests are stored at `enrollmentRequests/{tutorUid}/requests/{requestId}`. Tutors accept (selecting a time slot), decline, or request more information. Accepted enrollments are

written to `enrollments/{tutorUid}/roster/{enrollmentId}`. The `slotKey` format is `"day:slot"` (e.g., `mon:5-7`). Group batches are managed through a 3-step wizard with member selection from the live roster.

3.3.5 Messaging Module

In-app messaging uses a 1:1 conversation model. The `conversationId` is the sorted participant pair joined by `__`, ensuring both parties address the same thread. Messages include sender, text, timestamp, and read status. Real-time listeners provide instant delivery with read receipts.

3.3.6 Pro Subscription Module

Tutors upgrade to Pro through the `ProUpgradeScreen`. The eSewa payment flow uses a `WebView` that auto-submits an HMAC-SHA256-signed form. Verification performs four checks: HMAC validation, transaction ledger lookup (replay protection), amount/product cross-check, and server-to-server status API confirmation. Pro features include higher capacity ceilings, more active batches, a distinct Pro badge, a Featured section, and advanced analytics.

3.3.7 Admin Module

The admin dashboard provides platform statistics using `getCountFromServer`. The verification queue displays pending tutor submissions with document previews, AI pre-screen results, and approve/reject actions.

4. System Design

4.1 Requirement Specification

4.1.1 Functional Requirements

FR1: User Registration and Authentication

- Users can register as Student, Tutor, or Admin.
- Email/password and Google Sign-In authentication.
- Email verification before dashboard access.
- Role-based routing to dashboards.

FR2: Tutor Profile Management

- Multi-step profile with subjects, grades, pricing, location.
- Profile photograph and citizenship document upload.
- Availability toggle for search visibility.

FR3: Student Discovery and Filtering

- Map-based proximity search with Google Maps.
- Filters for subject, grade, budget, distance, tutoring mode.
- Distance-based ranking with Pro subscription priority.
- Saved tutors functionality.

FR4: AI-Powered Chatbot

- Natural-language queries about available tutors.
- RAG pipeline using Groq API with Firestore retrieval.
- Conversation history for follow-up questions.

FR5: Enrollment Request Handling

- Students send requests with preferred time slots.

- Tutors accept (with slot selection), decline, or request info.
- Enrollment tracking with start/end dates.

FR6: Group Batch Management

- Tutors create batches with name, subject, capacity, schedule.
- Member add/remove from live roster.
- Students browse open batches from enrolled tutors.

FR7: In-App Messaging

- 1:1 conversations with deterministic IDs.
- Read receipts and real-time delivery.

FR8: Ratings and Reviews

- Star ratings and text reviews for enrolled tutors.
- Running average computation.

FR9: AI Identity Verification

- Groq Vision OCR for citizenship document name extraction.
- Groq Vision face detection for profile photograph.
- Auto-approve at confidence ≥ 0.9 ; manual review otherwise.

FR10: Pro Subscription

- eSewa sandbox payment with HMAC-SHA256 signing.
- Enhanced capacity, batches, badges, and analytics for Pro.

FR11: Admin Dashboard

- Platform statistics with live aggregation.
- Verification queue with AI pre-screen results.
- User management.

4.1.2 Non-Functional Requirements

- **Usability:** Consistent NativeWind design system with tactile haptic feedback and accessible color tokens.

- **Performance:** Real-time Firestore listeners provide sub-second updates; map rendering and chatbot responses execute within acceptable time.
- **Security:** Firebase Security Rules enforce owner-only data access and role-based permissions.
- **Privacy:** Location data used solely for proximity calculations. Student locations masked from non-enrolled tutors.
- **Scalability:** Repository pattern with data source selector allows mock/production switching.
- **Maintainability:** Modular architecture with TypeScript strict mode and custom ESLint rules.

4.1.3 System Architecture

EdumentX follows a multi-layer hybrid architecture:

- **Presentation Layer:** React Native + NativeWind + Expo Router mobile application.
- **Application Layer:** TypeScript services with repository pattern. Screen components consume repository interfaces with runtime data source selection.
- **Service Layer:** Firebase Authentication, Cloud Firestore, Supabase Edge Functions (chat, eSewa, verify-identity), Supabase Storage.
- **Data Layer:** Cloud Firestore for structured data, Supabase Storage for files, Groq API for AI inference.

4.1.4 Use Case Diagram

The principal actors are Student, Tutor, and Admin. Student use cases include authentication, map-based discovery, AI chatbot, enrollment requests, batch browsing, messaging, saving tutors, and reviews. Tutor use cases include profile management, AI verification, enrollment handling, batch management, dashboard analytics, and Pro subscription. Admin use cases include verification review, platform statistics, and user management.

5. Results and Discussion

5.1 Software Development Approach

The project followed an iterative-incremental development model over approximately six months (February–August 2026). Each phase produced a working increment that was tested on a real Android device before proceeding. The mid-term defense (August 2026) covered the completed foundation: authentication, onboarding, design system, and admin panel. The post-mid-term phase added map-based discovery, AI verification, enrollment management, group batches, messaging, Pro subscription, and the RAG chatbot.

5.2 Testing and Evaluation

5.2.1 Unit Test Results

The enrollment logic test suite (`npm run test:derived`) covers 47 unit tests across all pure helper functions:

Table 5.1: Unit Test Coverage Summary

Module	Tests	Status
KTM date helpers	8	All passing
slotKey parsing	12	All passing
Today-sessions derivation	6	All passing
Booked-map generation	5	All passing
Capacity counting	8	All passing
Availability draft helpers	8	All passing
Total	47	All passing

5.2.2 Lint Rule Test Results

The design-token ESLint rule tests (`npm run test:lint-rules`) verify five custom rules:

Table 5.2: ESLint Rule Test Results

Rule	Tests	Status
no-raw-hex-placeholder	4	All passing
no-raw-hex-color-prop	4	All passing
no-raw-hex-inline-color	4	All passing
no-non-token-radius	5	All passing
no-non-token-color-class	4	All passing
Total	21	All passing

5.2.3 Security Rules Test Results

The Firestore security rules emulator suite (`npm run test:rules`) exercises six test suites:

Table 5.3: Security Rules Test Suites

Suite	Status
Tutors availability carve-out	Passing
Batches collections	Passing
AcceptRequest transaction paths	Passing
Conversations/messages participant gates	Passing
Reviews list rules	Passing
RemoveEnrollment cascade	Passing

5.2.4 Build Verification

The project passes TypeScript type checking (`npx tsc --noEmit`) with zero errors and ESLint (`npx eslint .`) with zero violations after every significant change. The Android debug build (`npx expo run:android`) completes successfully.

5.3 Current Build

Figure 5.1 shows the current build running on an Android device.



Figure 5.1: EdumentX Current Build on Android

6. Conclusions

This report presented the design, development, and evaluation of EdumentX, a location-based mobile application for home tutor discovery in Kathmandu Valley. The project successfully addresses the core problems of informal tutor finding, lack of verification, and absence of proximity-based search through a comprehensive mobile platform built with React Native and Expo.

The key achievements of this project are:

- A map-based tutor discovery system using Google Maps with custom pins and proximity-based ranking, enabling students to find nearby tutors within the Kathmandu Valley.
- An AI-powered RAG chatbot using Groq that answers natural-language queries about available tutors, grounded in live Firestore data.
- An automated identity verification pipeline using Groq Vision OCR for citizenship document validation and face detection for profile photographs, with automatic approval for high-confidence matches and manual admin review for ambiguous cases.
- A structured enrollment management system with slot-based scheduling, enrollment request workflows, and group batch management.
- An in-app messaging system with real-time delivery and read receipts.
- A Pro subscription model with eSewa payment integration (sandbox), providing enhanced visibility and features for subscribing tutors.
- A complete admin dashboard with verification queues, platform statistics, and user management.
- A zero-cost architecture operating within Firebase Blaze and Supabase free-tier quotas, making the platform viable as a demonstration prototype without paid infrastructure.

The project demonstrates that a location-first, trust-oriented tutoring marketplace can be built and deployed using modern cross-platform frameworks and serverless services at zero operational cost at demo scale.

7. Limitations and Future Enhancement

7.1 Limitations

1. **Android-only:** The current build targets Android exclusively. iOS support requires an Apple Developer account and additional testing with Apple Maps integration via expo-maps.
2. **eSewa sandbox:** The payment integration uses eSewa's sandbox environment with test credentials (EPAYTEST). Production deployment requires a registered merchant account and live credentials.
3. **No video tutoring:** The platform supports only in-person home tutoring. Online or video-based classes are outside the current scope.
4. **Limited tutor coverage:** The platform currently has a small number of registered tutors. A critical mass of tutors is needed before the recommendation engine and chatbot provide meaningful results.
5. **HuggingFace DNS:** The HuggingFace Serverless Inference API is unreachable from Supabase Edge Runtime due to DNS limitations. Face detection uses Groq Vision as a fallback.
6. **No push notifications:** While expo-notifications is installed, FCM push notifications for real-time alerts (messaging, enrollment updates) are not yet fully implemented. The current build uses local notifications only.

7.2 Future Enhancement

1. **iOS support:** Extend the build to iOS with Apple Maps integration and test on physical iOS devices.
2. **FCM push notifications:** Implement server-side push notifications via Firebase Cloud Functions for messaging, enrollment updates, and verification status changes.

3. **Online tutoring:** Add video call integration (e.g., WebRTC) for online tutoring sessions.
4. **Tutor analytics:** Expand the analytics dashboard with profile view trends, subject demand patterns, and conversion rates.
5. **Multi-language support:** Add Nepali language support for the UI and chatbot responses.
6. **EAS Build and distribution:** Set up EAS Build for automated Android builds and distribute via Firebase App Distribution for testing.

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